

(7.) $(7012)_8$ to Decimal & Hexadecimal,

7012

$$\Rightarrow 7 \times 8^3 + 0 \times 8^2 + 1 \times 8^1 + 2 \times 8^0$$

$$\Rightarrow (3594)_{10}$$

7	0	1	2
111	000	001	010

<u>111</u>	<u>000</u>	<u>001</u>	<u>010</u>
$\frac{8+4+2}{14}$	$\frac{0}{0}$	$\frac{8+2}{10}$	

$$\Rightarrow (EOA)_{16}$$

(5.) Hexadecimal to Decimal :-

$(101010111100)_2$

1010	1011	1100
$2^9 2^8 2^7 2^6$	$2^5 2^4 2^3 2^2$	$2^1 2^0 2^{-1} 2^{-2}$
8+2	8+2+1	8+4
10	11	12
A	B	C

$\Rightarrow (ABC)_{16}$

101010111100
 $2^9 2^8 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0$
 $\Rightarrow 2^9 + 2^8 + 2^7 + 2^5 + 2^4 + 2^3 + 2^2$
 $\Rightarrow (2748)_{10}$

(6.) $(7654)_8$ to Decimal & Hexadecimal

$7 \quad 6 \quad 5 \quad 4$
 $\Rightarrow 7 \times 8^3 + 6 \times 8^2 + 5 \times 8^1 + 4 \times 8^0$
 $\Rightarrow (4012)_2$

$7 \quad 6 \quad 5 \quad 4$
 $111 \quad 110 \quad 101 \quad 100$

111	110	101	100
$8+4+2+1$	$8+2$	$8+4$	
15	10	12	

15 10 12

$(FAC)_{16}$

$$1.) (110110101101)_2$$

$$\Rightarrow \begin{array}{cccccccccccc} 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 2^0 & 2^1 & 2^2 & 2^3 & 2^4 & 2^5 & 2^6 & 2^7 & 2^8 & 2^9 & 2^{10} & 2^{11} \end{array}$$

$\Rightarrow$

$$2648 + 1024 + 256 + 128 + 32 + 8 + 4 + 1$$

$$\Rightarrow (3501)_{10}$$

$$\begin{array}{ccccccc} 110 & 110 & 101 & & & & \\ 2^2 2^1 2^0 & 2^2 2^1 2^0 & 2^2 2^1 2^0 & & & & \\ 4+2 & 4+2 & 4+0+1 & & & & \\ 6 & 6 & 5 & & & & \end{array}$$

$$\Rightarrow (6655)_8$$

(i.) Hexadecimal to Binary: $(12345)_{16}$

2	1	2	3	4	5	1
2	6	1	7	2	0	
2	3	0	8	8	2	0
2	1	5	4	3	1	
2	7	7	1	1		
2	3	8	5	1		
2	1	4	2	0		
2	9	6	0			
2	4	8	0			
2	2	4	0			
2	1	2	0			
2	6	0				
2	3	1				
						1

$$(12345)_{16} = (110000000111001)_2$$

16	1	2	3	4	5	9
16	7	7	1	3		
16	4	8	0			
						3

$$(12345)_{16} = (3039)_{16}$$

(2.) Hexadecimal & Binary, $(65535)_{16}$

$$\begin{array}{r|l}
 16 & 65535 \quad 15 \\
 \hline
 16 & 4095 \quad 15 \\
 \hline
 16 & 255 \quad 15 \\
 \hline
 & 15
 \end{array}$$

$\Rightarrow (FFFF)_{16}$

$$\begin{array}{r|l}
 2 & 65535 \quad 1 \\
 \hline
 2 & 32767 \quad 1 \\
 \hline
 2 & 16383 \quad 1 \\
 \hline
 2 & 8191 \quad 1 \\
 \hline
 2 & 4095 \quad 1 \\
 \hline
 2 & 2047 \quad 1 \\
 \hline
 2 & 1023 \quad 1 \\
 \hline
 2 & 511 \quad 1 \\
 \hline
 2 & 255 \quad 1 \\
 \hline
 2 & 127 \quad 1 \\
 \hline
 2 & 63 \quad 1 \\
 \hline
 2 & 31 \quad 1 \\
 \hline
 2 & 15 \quad 1 \\
 \hline
 2 & 7 \quad 1 \\
 \hline
 2 & 3 \quad 1 \\
 \hline
 & 1
 \end{array}$$

$\Rightarrow (1111111111111111)_{2}$

(1.) Binary
Octal

8	1023.7
8	127.7
8	15.7
	1

$$(1023)_{10} = (1777)_8$$

2	10231
2	5111
2	2551
2	1271
2	631
2	311
2	151
2	71
2	31
2	1

$$\Rightarrow (1111111111)_2$$

(13.) to Hexadecimal & Decimal

$$(111100001111)_2$$

$$\frac{1111}{8+4+2+1} \quad \frac{0000}{0} \quad \frac{1111}{8+4+2+1}$$

$$\frac{15}{15}$$

$$\Rightarrow (15015)_{16} \Rightarrow (FOF)_{16}$$

$$\Rightarrow 2^{10} 2^9 2^8 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0$$

$$\Rightarrow 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$$

$$\Rightarrow (3855)_{10}$$

(14.) to Decimal and Octal

$$(101111001010)_2$$

$$\Rightarrow 2^{10} 2^9 2^8 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0$$

$$\Rightarrow 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$$

$$\Rightarrow (3018)_{10}$$

$$\frac{101}{4+1} \quad \frac{111}{4+2+1} \quad \frac{001}{1} \quad \frac{010}{2}$$

$$\frac{5}{5}$$

$$\Rightarrow (5712)_8$$

(12.) $(4095)_{10}$ to Binary $\&$ Octal

$$\begin{array}{r} 2 \overline{) 4095} \\ 2 \overline{) 2047} \\ 2 \overline{) 1023} \\ 2 \overline{) 511} \\ 2 \overline{) 255} \\ 2 \overline{) 127} \\ 2 \overline{) 63} \\ 2 \overline{) 31} \\ 2 \overline{) 15} \\ 2 \overline{) 7} \\ 2 \overline{) 3} \\ 2 \overline{) 1} \end{array}$$

$$\Rightarrow (111111111111111)_2$$

$$\frac{111}{7} \quad \frac{111}{7} \quad \frac{111}{7} \quad \frac{111}{7}$$

$$\Rightarrow (7777)_8$$

(13.) ~~Dec~~

$$\begin{array}{r} 8 \overline{) 4095} \\ 8 \overline{) 511} \\ 8 \overline{) 63} \\ 7 \end{array}$$

$$\Rightarrow (7777)_8$$

(11.) $(703)_{16}$ to Binary ~~to a decimal~~

$$\begin{array}{r} 7 \quad 0 \quad 3 \\ \hline 0111 \quad 1101 \quad 0011 \end{array}$$

$$\Rightarrow (011111010011)_2$$

$$\begin{array}{r} 011 \quad 111 \quad 010 \quad 011 \\ \hline 2^{+1} \quad 4^{+2+1} \quad 2 \quad 2^{+1} \\ \hline 3 \quad 7 \quad 2 \quad 3 \end{array}$$

$$\Rightarrow (3723)_8$$

$$\begin{aligned} &\Rightarrow 011111010011 \\ &\quad 2^{11} 2^{10} 2^9 2^8 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0 \\ &\Rightarrow 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^1 + 2^0 \\ &\Rightarrow (2003)_{10} \end{aligned}$$

(10.)

Binary to Octal
(1500)₁₀

2	1500	0
2	750	0
2	375	1
2	187	1
2	93	1
2	46	0
2	23	1
2	11	1
2	5	1
2	2	0
1		

$\Rightarrow (10111011100)_2$

010	11	011	100
2	7	3	4

(8.) Binary to Octal

(2FA3)

2	F	A	3
0010	1111	1010	0011

$\Rightarrow (0010111110100011)_2$

010	111	110	100	011
$\frac{2}{2}$	$\frac{7+2+1}{7}$	$\frac{4+2}{4}$	$\frac{4}{4}$	$\frac{2+1}{3}$
2	7	6	4	3

$\Rightarrow (27643)_8$

(9.) $(B4F9)_{16}$ to Binary to Octal

B	4	F	9
1000	0100	1111	1001
1011			

$\Rightarrow (1011010011111001)_2$

1001	011	010	011	111	001
$\frac{1}{3}$	$\frac{3}{2}$	$\frac{2}{3}$	$\frac{7}{7}$	$\frac{1}{7}$	$\frac{1}{1}$

$\Rightarrow (132371)_8$